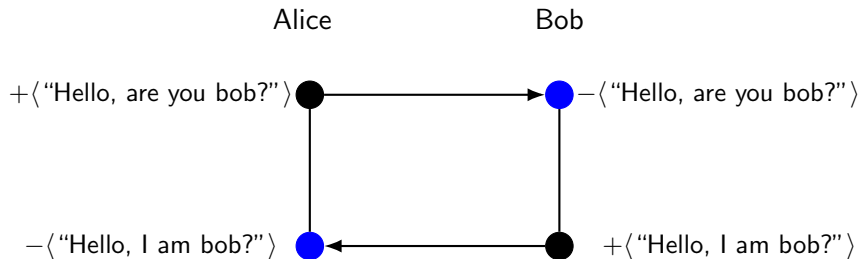


Session 3: CPSA Etymology and Attacks in CPSA

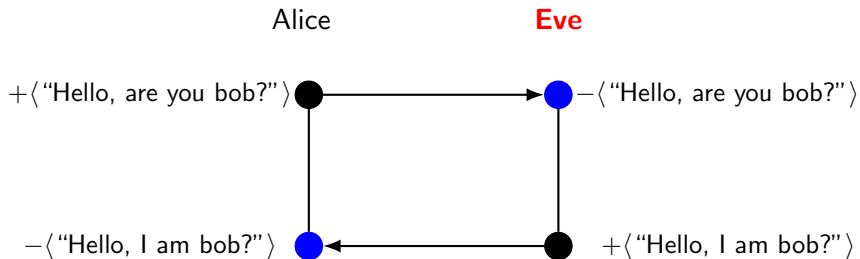
Slides prepared by UMBC Protocol Analysis Lab

Last edited July 28, 2026

A completed protocol?



Strands can lie



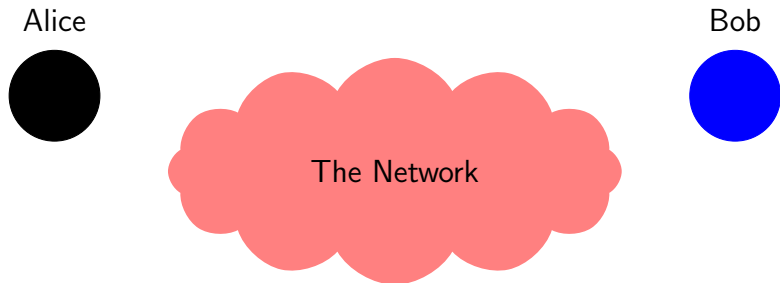
CPSA adversarial strand actions

CPSA adversarial strand actions

- M. Create Primitive: $\langle +p \rangle$ where $p \in P$ where P is the set of all basic values.
 - C. Concatenation: $\langle -g, -h, +gh \rangle$ where $g, h, gh \in G$
 - S. Separation into components: $\langle -gh, +g, +h \rangle$
 - E. Encryption $\langle -K, -h, +\{h\}_K \rangle$
 - D. Decryption $\langle -K, -\{h\}_K, +h \rangle$
- 1. $\langle +m \rangle$ emit a message from the network.
 - 2. $\langle -m \rangle$ extract a message from the network.
 - 3. $\langle m_1 m_2 \rangle$ concatenate two messages.
 - 4. $\langle \{m\}_K \rangle$ encrypt a message m under key K .

An adversarial model of the network

An adversarial model of the network

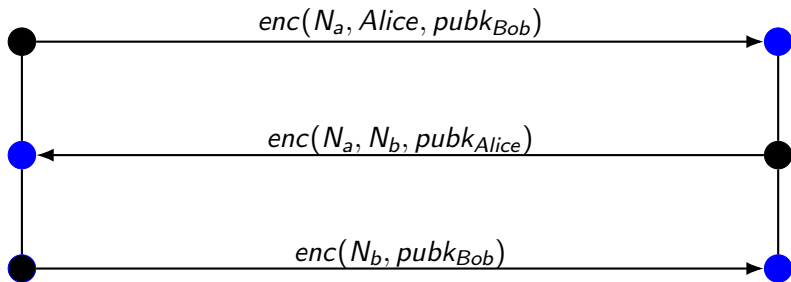


Goal: Mutually *authenticate* (verify identities) over an insecure network.

Needham Schroeder

Alice

Bob



Is this protocol secure against the Dolev-Yao Adversary?

● = Reception ● = Transmission

How can we spot attacks?

How can we spot attacks?

► *SECURITY GOALS*

How can we spot attacks?

- ▶ *SECURITY GOALS*
- ▶ Dotted lines

How can we spot attacks?

- ▶ *SECURITY GOALS*
- ▶ Dotted lines
- ▶ Listener Strands

How can we spot attacks?

- ▶ *SECURITY GOALS*
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- ▶ Missing/extra strands

How can we spot attacks?

- ▶ *SECURITY GOALS*
 - ▶ Dotted lines
 - ▶ Listener Strands
 - ▶ Missing/extra strands
-
- ▶ What do we notice in the NS shapes from last time?

NS shapes from last time

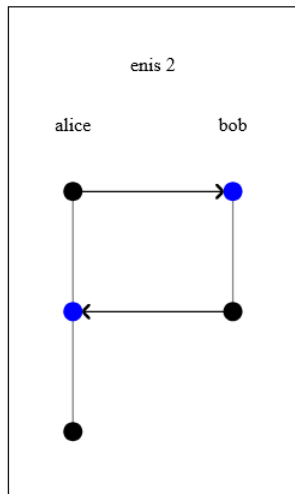


Figure: NS from Alice's perspective

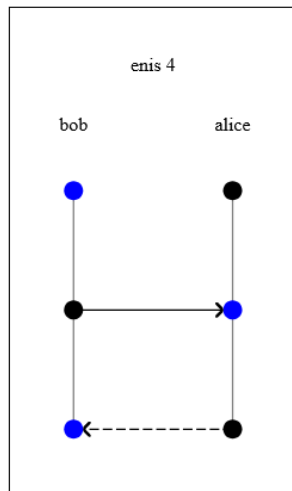
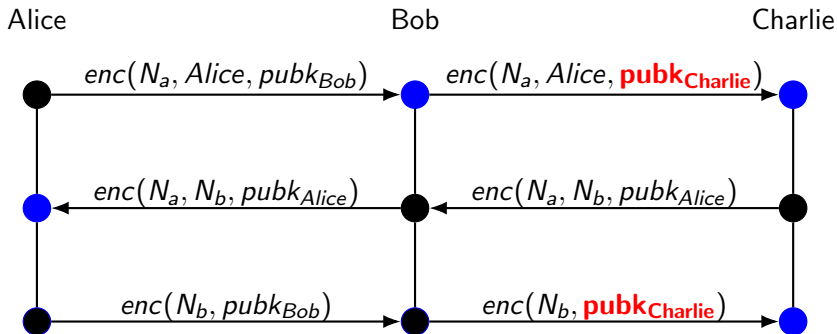


Figure: NS from Bob's perspective

Needham Schroeder Attack

Needham Schroeder Attack



Remember: The Dolev-Yao Adversary can be a legitimate party on the network

● = Reception ● = Transmission

What do we give CPSA?

`protocol.scm`

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`protocol.scm`

▶ `herald`

What do we give CPSA?

`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`

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`protocol.scm`

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 - ▶ `defrole`

What do we give CPSA?

`protocol.scm`

- ▶ `herald`
- ▶ `defmacro`
- ▶ `defprotocol`
 - ▶ `defrole`
- ▶ `defskeleton`
- ▶ `defstrand`

What does CPSA give us?

`protocol.scm` →

What does CPSA give us?

protocol.scm →

▶ protocol.txt

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`
- ▶ `protocol.xhtml`

What does CPSA give us?

`protocol.scm` →

- ▶ `protocol.txt`
- ▶ `protocol_shapes.txt`
- ▶ `protocol.xhtml`
- ▶ `protocol_shapes.xhtml`

Homework

Find attacks in CPSA

Find attacks in CPSA

Looking at shapes to find attacks takes some practice, so let us do some practice!

Two of the protocols are already done, you just have to run make so CPSA can make shapes, and then analyze them.

Two of the protocols need to be filled in, you will have to add part of a role, fill in a skeleton, or something else (documented in the comments), and then run CPSA to analyze the shapes.

Survey

Due at 6:00pm on Thursday the 23rd

